Morphology-Conditioned Physics-Based Motion Imitation Across 128 Body Shapes

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Abstract

Physics-based motion imitation must accommodate both the diversity of human movement and differences in body geometry and dynamics. We construct a text-associated, multi-morphology motion corpus from AMASS-derived HumanML3D clips using HUMOS, pairing shape-specific reference motions with matching SMPL-based simulation assets. The current corpus contains 20,951 clip identities, including mirrored clips, across 128 body configurations, yielding 2,681,728 generated variants. Source captions remain associated metadata rather than controller inputs. A contact-aware kinematic refinement procedure reduces selected reference artifacts while retaining explicit limits on pose and root changes. We train one morphology-conditioned policy directly on the refined multi-shape references, using temporal slot/type attention and a streamed training pool. Deterministic, clip-grouped splits and morphology-matched evaluation distinguish held-out motion tracking from training-pool monitoring. Offline evaluation of the selected checkpoint reports 98.47% success on the 1,048-clip validation split and 99.14% on the independent 1,048-clip test split, each with one selected body configuration per clip and a capped horizon; these are trained-body, one-shape-per-clip results, not all-shape or unseen-body tests. A corrected eight-run grid of four temporal architectures crossed with refined and unrefined references, on a 150-clip development set, selects temporal slot/type attention without adaLN-Zero. Residual failures on both splits are analysed against inherited captions and concentrate on ground-contact transitions and whole-body acrobatics. An unseen-morphology generalisation study within the trained coefficient range, and an analysis of how body shape and motion type jointly set control demand, are specified but not yet run.

1 Introduction

A motion-imitation controller should reproduce an action without assuming that every character has the same body. Changes in limb lengths, mass distribution, and collision geometry alter both the reference trajectory and the dynamics required to follow it. Replaying an identical joint sequence across bodies does not necessarily preserve foot placement, clearance, or balance. Conversely, optimizing only spatial agreement can obscure whether the intended articulation has been retained. Learning across body shapes therefore requires a consistent relationship between source motion, shape-specific reference, and simulated body.

Existing work provides large motion-capture collections, body-shape-conditioned motion generation, and scalable physics-based control [1, 2, 3, 4, 5, 6]. We build on these components to study a practical question: how can one policy imitate a large collection of shape-specific motions while preserving traceable data identities and a meaningful evaluation protocol? Our focus is their integration and empirical assessment, not a claim that morphology conditioning or attention is new.

Our data pipeline extends AMASS-derived HumanML3D clips to 128 body configurations. HUMOS generates the corresponding reference motions; SMPLSim constructs simulation assets from the same body parameters. A conservative refinement pass smooths rotations, reduces stance sliding through root translation, and enforces clearance of the modeled collision geometry. The inherited descriptions connect each source action to its generated variants. This association supports paired analysis and future language-based uses, but the controller studied here consumes reference motion, not text.

The selected controller combines current state, dilated state history, and future reference targets through temporal slot/type attention. Body parameters and a previous action are concatenated into separate actor and critic heads. The main experiment trains directly on multiple shapes; neutral-body pretraining and mixed-source reward training are not part of the final method. We use a 150-clip development subset to make design exploration affordable, followed by full-scale training on a versioned split. Corrected small-scale reruns are now reassessing the design choices made during that development.

This draft makes three contributions:

1. A derived, text-associated multi-morphology corpus, with an explicit construction procedure linking source clips, generated references, body assets, and reproducible splits. Its captions are inherited rather than newly annotated for each body.

2. A single morphology-conditioned imitation system operating across 18,855 training clips and 128 body configurations, integrating temporal observations, matching simulation assets, reference refinement, and memory-bounded motion streaming.

3. An empirical protocol separating architecture, reference quality, and evaluation correctness: an eight-run architecture/reference grid on a 150-clip development set, a full-scale validation and independent test evaluation of the selected checkpoint, and a caption-grounded analysis of the residual failures.

The selected checkpoint reaches 98.47% success on validation and 99.14% on the untouched test split under the protocol in Section 5.3. These results concern held-out clip identities on trained body configurations, with one body evaluated per clip; they do not establish success for every shape or for unseen bodies. Section 6 also specifies, without results, an evaluation on body shapes drawn from the trained coefficient range but excluded from the 128 training configurations (Section 6.5.1), and a study of how morphology and motion type jointly shape physical control demand (Section 7).

2 Related Work

2.1 Physics-based motion imitation

DeepMimic demonstrated reference-guided reinforcement learning for simulated character skills [4]. Perpetual Humanoid Control developed robust humanoid imitation for real-time simulated avatars [5]. These approaches motivate learning a feedback controller rather than directly replaying a kinematic trajectory. Subsequent single-body trackers scaled this feedback-control approach in coverage and deployability: PULSE distills a low-dimensional, high-coverage motion representation for general humanoid control [14]; OmniH2O and ExBody2 target dexterous, deployable whole-body teleoperation and expressive control on physical humanoids [15, 16]; GMT uses a motion mixture-of-experts to track a broad motion distribution with one policy [17]. Each of these fixes the body and scales the motion distribution; we instead fix the tracking task and scale the body, holding 128 configurations under one policy. We implement our experiments in ProtoMotions [6] with Isaac Gym [7], using PPO optimization [8]. Our contribution is the multi-morphology data and control workflow and its evaluation, rather than a replacement for these simulation or learning frameworks.

2.2 Morphology-conditioned control and retargeting

Body-shape variation has already been studied in physics-based control: Won and Lee learned controllers that accommodate variation in character shape [9]. Morphology conditioning is therefore an established idea, and our work should not be interpreted as introducing it. We instead examine one controller over a large set of paired motions and bodies, with shape identity maintained throughout data generation, simulation, and evaluation.

Kinematic retargeting and dynamic tracking address different requirements. SMPL provides a common parameterized body representation [10], and SMPLSim constructs simulation-ready assets from these models [11]. A shared skeleton makes poses comparable, but does not make global positions or required actuation independent of shape. Matching a reference to its intended simulation asset is consequently part of the task definition, not an optional evaluation convenience.

2.3 Text-motion data, shape-conditioned generation, and refinement

AMASS consolidates motion capture in a common body-model representation [1]. HumanML3D associates processed motion segments with natural-language descriptions [2]. We inherit those source identities and descriptions rather than claiming new motion capture or shape-specific language annotation.

HUMOS generates identity-conditioned human motion using a cycle-consistent learning formulation [3]. We use its pretrained model to obtain shape-specific references, not as the physics controller and not as a text-conditioned policy. The resulting sequences still require conversion to the simulator's skeleton and coordinate conventions, and their feasibility must be assessed in that simulator. Our refinement is a deliberately limited kinematic post-process; it is not a new generative model or a full dynamic trajectory optimizer.

The controller uses standard Transformer operations [12]. Learned slot and type embeddings identify temporal positions and observation sources. An ablation also investigates actor-only morphology modulation with adaLN-Zero, adapting the zero-initialized conditioning design used in Diffusion Transformers [13]. That experiment tests a conditioning mechanism, not diffusion-based action generation.

3 Multi-Morphology Text-Associated Motion Dataset

We construct a derived corpus in which a source clip identity connects a motion description, 128 generated motion variants, and their matching simulation assets. A *clip* denotes a source segment, including a mirrored segment when present; a *variant* denotes that clip on one body configuration. These units must not be conflated when reporting dataset size or evaluation coverage. Figure 1 shows the construction paths.

3.1 Source motion and text provenance

We use the AMASS-derived motion portion of HumanML3D [1, 2], with the 18 source collections listed in Appendix A.1. Source SMPL+H sequences are processed at 20 Hz, including the Z-up to Y-up coordinate conversion used by the HumanML3D/HUMOS preprocessing pipeline. Clip

Motion	AMASS $\longrightarrow$ HumanML3D preprocessing and clip extraction $\longrightarrow$ HUMOS conditioned on $(g, \boldsymbol{\beta})$ $\longrightarrow$ MotionLib conversion and reference refinement $\longrightarrow$ per-clip motion variants.
Body	SMPL with the same $(g, \boldsymbol{\beta})$ $\longrightarrow$ SMPLSim $\longrightarrow$ matching skeleton, collision geometry, mass/inertia, and MJCF asset identity.
Text	HumanML3D source descriptions $\longrightarrow$ clip-ID lookup associated with the generated variants (metadata only; not a policy input).

The body assets are used for motion conversion, shape-specific forward kinematics, ground-clearance correction, and simulation. Refined reference motion and matching simulated state feed one shared policy.

Figure 1: Dataset and asset construction. The body parameters are shared across the motion and body paths; text follows a separate identity-association path.

extraction, temporal cropping, and mirroring preserve the HumanML3D identities. The preparation also includes treadmill/skating cleanup. This subset is not the entirety of AMASS.

The generation pipeline produced 22,459 candidate clip identities, counting original and mirrored clips. A subsequent text-based filter excluded 1,508 clips identified as requiring external objects or environmental support, leaving 20,951 identities for the flat-ground imitation task. This filter is a task-scope heuristic: it does not prove that every retained reference is support-free or dynamically feasible. The retained and excluded ID lists provide the audit trail for that decision.

Source descriptions are available through a separate clip-ID lookup and the HumanML3D annotation metadata used by HUMOS. A description is associated with the variants of its source clip; it is not a new human annotation for each generated body. Coverage, segment timing, and semantic fidelity of this association have not yet been exhaustively audited. In particular, mirrored motions may require left/right wording changes, and simply removing a mirror prefix does not establish a correct caption.

3.2 Body sampling and shape-specific generation

Let a body configuration be $b = (g, \boldsymbol{\beta})$, where g identifies the female or male SMPL body-model variant and $\boldsymbol{\beta} \in \mathbb{R}^{10}$ contains shape coefficients. We sample

$$\beta_{ij} \sim \mathcal{U}(-3, 3), \qquad i = 1, \dots, 64, \quad j = 1, \dots, 10, \tag{1}$$

using NumPy's `default_rng(46)`, and cross these 64 vectors with the two body-model variants. Thus the corpus contains 128 body configurations, not 128 independently sampled beta vectors. The actual coefficient arrays and stable beta keys identify the bodies; the keys are not content hashes. Uniform sampling of model coefficients is a controlled source of variation, not a representative sample of human populations.

For each source clip, we provide the processed pose/root motion features and the target body parameters to the pretrained HUMOS model [3]. The local configuration uses root-relative 6D root and joint rotations together with translation, with shape and body-model conditioning. The generated pose and root sequences are retained for conversion. HUMOS is an existing identity-conditioned model; neither its training nor a text-to-motion generator is a contribution of this work. The checkpoint used in the documented pipeline is identified in Appendix A.4.

In parallel, SMPL bodies [10] with identical parameters are converted to MJCF assets using SMPLSim [11]. The asset-generation configuration uses mass estimation and the configured hand/foot simplifications (`real_weight`, `replace_feet`, `remove_toe`, and `freeze_hand`). The resulting imitation representation contains 24 rigid bodies and 69 actuated joint coordinates. Collision proxies, joint ranges, and estimated masses/inertias belong to each asset. These physical parameters are simulation-model estimates, not measurements of corresponding human participants.

3.3 Motion conversion and reference refinement

The generated sequences are converted to ProtoMotions' Z-up simulation convention and resampled from 20 to 30 Hz. Forward kinematics (FK) uses the matching asset rather than a single neutral skeleton. The packaged representation includes global body positions and rotations, local rotations, joint positions, linear/angular and joint velocities, and contact labels. The earlier corpus already included frame-zero grounding; the newer refinement addresses artifacts throughout each sequence.

Refinement is applied independently to each variant in the following order. First, stored ankle/toe contacts are validated, or contacts are detected from foot position and velocity if usable labels are unavailable. Short gaps are closed and very short contact runs are removed. These labels define candidate stance intervals; they are not measured ground-reaction forces.

Second, joint rotations are temporally filtered using sign-continuous quaternion representations, normalization, and a bounded geodesic correction. Contact-transition frames divide the sequence into segments so filtering does not average across a change of stance. A transition taper and a limited smoothing blend further restrict changes near those boundaries. Post-smoothing FK then recomputes the positions used by subsequent corrections.

Third, a conservative root-translation correction reduces horizontal stance sliding. For each contact interval, an anchor is formed from the foot's median horizontal position. A least-squares correction seeks a common root XY shift that brings the active feet closer to their anchors; the shift is temporally regularized and capped. Multiple contacts can disagree, so this root-only operation does not guarantee exact foot locking and does not solve for new joint angles.

Fourth, we evaluate the lowest point of the matching collision geometry at every frame. With floor height zero and margin ϵ_z, the required lift is

$$\ell_t^{\mathrm{req}} = \max\left(0, \epsilon_z - \min_{x \in \mathcal{C}_b(t)} x_z\right), \tag{2}$$

where $\mathcal{C}_b(t)$ denotes the modeled collision geometry for body b. A smoothed lift envelope is constrained to remain above this requirement. This prevents penetration of the checked geometry in the reference sequence; it does not establish balanced support, realistic impacts, or freedom from self-collision.

Finally, FK and velocities are recomputed from the corrected root and rotations, independently within each motion. This avoids inconsistent position/velocity tensors and artificial derivatives across clip boundaries. The default thresholds, filter windows, and correction bounds are given in Table 11.

The refinement objective is modest: reduce visible and measurable kinematic artifacts while preserving the generated motion. Before/after assessment must therefore include both artifact measures (sliding, penetration, and jerk) and fidelity measures (rotation/root changes and retained cross-shape variation). Improved kinematics alone does not demonstrate improved policy learning or dynamic feasibility.

3.4 Packaging, scale, and deterministic splits

Each per-clip file contains the 128 variants, their frame counts and sampling rates, and the clip/asset identities needed to recover the corresponding body parameters. Captions remain a separately associated metadata resource; they are not established as a field of every training tensor file. This

Table 1: Current corpus and split sizes. Variants are generated combinations, not independent captures. Every clip has 128 body configurations.

Partition	Clip identities	Motion variants
Training	18,855	2,413,440
Validation	1,048	134,144
Test	1,048	134,144
Total	20,951	2,681,728

packaging allows the training system to stream clip files without downloading the complete corpus at startup.

We use the versioned `hhi_stage2_v1` split with seed 42 and approximate 90/5/5 proportions (Table 1). All variants of a clip stay together, and original/mirrored partners are assigned as a group. The 150 development clips are reserved for training, together with their mirror partners. Training accepts explicit training and validation manifests; test-manifest membership is not loaded by the training workflow. Split hashes and version information accompany the run and checkpoints.

These splits test held-out clip identities on the trained body set. They have not been demonstrated to be disjoint by subject or original capture recording: different segments can still share a source recording. This distinction limits claims about broader motion generalization.

3.5 Current resource scope

The current resource combines generated references, matching assets, inherited text associations, and reproducible splits. It is not a fully audited text-and-motion benchmark release. Caption completeness and alignment, body-specific language, licensing and redistribution conditions, and shape-conditioned semantic fidelity require further work. Those extensions can be developed independently of the reference-conditioned controller evaluated here.

4 Morphology-Conditioned Motion Imitation

4.1 Task, simulation, and actuation

For a clip c and body configuration b, let $\hat{s}_{c,b}(t)$ be the refined reference state and s_t the simulated state of the matching asset. We learn a shared policy $\pi_\theta(a_t \mid o_t, b)$ that tracks this reference on flat ground. The body assigned to an environment is fixed, while its reference motion can change on reset. No language input or external scene object is supplied.

Experiments use the `smpl_mor` assets in Isaac Gym through ProtoMotions [7, 6]. Simulation runs at 60 Hz with two substeps and control decimation two, giving a 30 Hz policy rate. The policy outputs a raw action $a_t \in \mathbb{R}^D$, where $D = 69$. Joint targets are

$$q_t^{\text{target}} = q^{\text{offset}} + q^{\text{scale}} \odot \tanh(a_t), \tag{3}$$

with offsets and scales derived from the configured joint ranges. These targets drive the simulator's built-in position PD controller. The action is not a residual added to the reference pose.

PD stiffness, damping, and effort limits are scaled per joint by the mass of its associated rigid body relative to that body in the reference asset. This is a body-level mass scaling, not a single scale based on total character mass, and it does not guarantee dynamically equivalent actuation across geometrically different bodies. Base gains and limits are listed in Table 12.

Table 2: Observation inputs supplied to both actor and critic. Dimensions are per slot. Positions/heights use meters, linear velocities m/s, angular velocities rad/s, and 6D rotations are dimensionless. Raw actions and morphology coefficients are not joint-angle measurements.

Input	Dimension	Slots	Representation and path
Current state	$15B - 2 = 358$	1	Height; $3(B-1)$ positions; $6B$ rotations; $3B$ linear and $3B$ angular velocities. Current-state encoder.
History	358	7	Same state representation at the seven past offsets. Shared history encoder.
Future reference	$24B = 576$	4	Target and relative poses ($18B$), plus velocity differences ($6B$). Shared future encoder.
Historical action	$D = 69$	1	A raw previous policy-action vector, before tanh and PD scaling. Bypasses attention.
Morphology	11	1	$[g, \beta_1, \ldots, \beta_{10}]$ from the assigned asset. Bypasses attention.

4.2 Observation design

The observation combines present state, history, future reference targets, a historical action, and explicit morphology information. Let $B = 24$ denote the number of simulated bodies. The state representation contains root height above ground, root-relative body positions excluding the root, 6D body rotations, and linear/angular velocities. Positions and velocities use a heading-aligned frame, and rotations remove the root heading. This removes dependence on arbitrary global heading and horizontal origin while retaining height and body configuration.

We include seven historical states at control-step offsets $\{1, 2, 3, 4, 8, 16, 32\}$, covering approximately 1.07 s, and four future reference slots at $\{1, 2, 4, 8\}$, covering approximately 0.27 s. Each historical state uses its own root/heading frame. Future slots contain target positions relative to the current root, heading-aligned target rotations, position differences from the current body state, relative rotations $R_{t,j}^{-1} \hat{R}_{t+k,j}$, and heading-aligned linear/angular velocity differences. Thus relative rotations are expressed with respect to the current body rotation, not as another heading-frame position feature.

The temporal context exposes both recent changes and more distant motion information; the future targets provide a short anticipation window. These are design motivations, not established ablation outcomes. History is initialized consistently with reference-state resets using available historical reference states; single-state initialization is the fallback when a trajectory history is unavailable. Contact flags, an explicit phase scalar, text, and estimated physical feature vectors are not part of the selected observation.

4.3 Temporal slot/type attention

We use three observation encoders: one for the current state, one shared across history slots, and one shared across future slots. Each is an MLP with two 256-unit ReLU hidden layers and a 256-dimensional token output. Inputs use running mean/standard-deviation normalization and a magnitude clamp of five.

The resulting 12 tokens comprise one current, seven historical, and four future tokens. For slot i, the Transformer input is

$$z_i = E_{\tau(i)}(x_i) + e_i^{\text{slot}} + e_{\tau(i)}^{\text{type}}, \qquad i = 0, \ldots, 11, \tag{4}$$

where $\tau(i)$ identifies the current/history/future source. Slot embeddings distinguish the configured temporal offsets, while type embeddings distinguish the observation sources. The Transformer has

Current state: 1×358 $\longrightarrow$ Current encoder: 1×256 token
Past states: 7×358 $\longrightarrow$ History encoder: 7×256 tokens
Future targets: 4×576 $\longrightarrow$ Future encoder: 4×256 tokens

$\downarrow$
12 tokens + learned temporal-slot and source-type embeddings
$\downarrow$
2 Transformer layers, 4 heads, width 256
$\downarrow$
Pool current token: $h_t \in \mathbb{R}^{256}$
$\downarrow$
Concatenate $[h_t,\ a_t^{\text{hist}},\ g, \boldsymbol{\beta}]$ and normalize

Actor: six 2,896-unit layers $\longrightarrow$ 69 action means $\longrightarrow$ tanh/range scaling $\longrightarrow$ PD targets.

Critic: four 1,024-unit layers $\longrightarrow$ scalar state value.

The entire encoder/attention pathway is instantiated separately for actor and critic; the drawing summarizes the common design, not shared weights. Morphology and action bypass attention into each head. Text is absent.

Figure 2: Observation-to-action data flow of the selected slot/type policy. Both networks receive the same input categories but have separate parameters.

two layers, four attention heads, token width 256, and feed-forward width 1,024 [12]. Its current-state token supplies the pooled output after the configured output projection and activation.

The final head input is the concatenation of the 256-dimensional attention output, the historical raw action, and the 11-dimensional morphology vector. This 336-dimensional input also uses running normalization and clamp five. Betas are not manually divided by three, embedded as a separate attention token, or used for adaLN modulation in the selected model.

Actor and critic have separate token encoders, attention modules, slot/type embeddings, and MLP heads. The actor head has six hidden layers of width 2,896; the critic has four of width 1,024, all with ReLU activations. The outputs are a Gaussian action mean and a scalar value, respectively. PPO samples actions with fixed diagonal log standard deviation -2.9; evaluation uses the mean action. Figure 2 summarizes the flow.

4.4 Tracking reward, resets, and PPO

The training reward combines global body tracking and root-height tracking. For corresponding simulated and reference body positions $p_j, \hat{p}_j$, rotations $R_j, \hat{R}_j$, and velocities $v_j, \hat{v}_j$ and $\omega_j, \hat{\omega}_j$, define

$$e_p = \frac{1}{3B}\sum_{j=1}^{B}\|p_j - \hat{p}_j\|_2^2, \qquad e_R = \frac{1}{B}\sum_{j=1}^{B} d_{\mathrm{SO}(3)}(R_j, \hat{R}_j)^2, \tag{5}$$

$$e_v = \frac{1}{3B}\sum_{j=1}^{B}\|v_j - \hat{v}_j\|_2^2, \qquad e_\omega = \frac{1}{3B}\sum_{j=1}^{B}\|\omega_j - \hat{\omega}_j\|_2^2, \tag{6}$$

$$e_h = (h - \hat{h})^2, \tag{7}$$

where $d_{\mathrm{SO}(3)}$ is the geodesic rotation angle in radians. The coordinate-wise means in the position and velocity errors match the implemented reductions. The reward is

$$\begin{aligned} r_t = 0.5\exp(-25e_p) + 0.3\exp(-5e_R) + 0.1\exp(-0.5e_v) \\ + 0.2\exp(-0.1e_\omega) + 0.2\exp(-100e_h). \end{aligned} \tag{8}$$

The weights are not normalized to sum to one. There is no explicit effort, action-smoothness, or contact-matching reward in this experiment, and there is no source-dependent switch between canonical and HUMOS reward functions.

References are sampled only from variants matching the environment's asset. Reference-state initialization starts at frame zero with probability 0.2; otherwise it uses a random valid reference time. Training episodes have a maximum of 1,000 control steps and use the configured fall termination with a 0.15 m height setting and permitted foot-contact bodies. Tracking-error termination is disabled. This training rule is distinct from the evaluation failure threshold in Section 5.4.

We optimize using PPO [8], with 32-step rollouts, discount 0.99, GAE parameter 0.95, policy-ratio clipping 0.2, and normalized advantages. Separate Adam optimizers use learning rates 2×10^{-5} for the actor and 10^{-4} for the critic. Other configuration values and actual run budgets appear in Table 3. Evaluation rollouts use deterministic inference without gradient tracking.

4.5 Direct multi-shape learning and streamed sampling

The final approach is direct multi-shape training on refined HUMOS references. A small static corpus supports initial development; the full-scale experiment streams per-clip files through `GlobalClipPool`. The training manifest is deterministically partitioned across GPU ranks. Each rank maintains a bounded resident pool containing all 128 variants of its selected clips. Environment-level motion sampling respects the assigned morphology.

At each scheduled rebuild, most resident slots are selected by explicit top priority, combining estimated clip difficulty and an exploration bonus. A fraction 0.2 is reserved for uniform random selection, maintaining rehearsal outside the highest-priority set. Difficulty estimates are updated from training-pool evaluations with an exponential moving average of rate 0.1 and a floor of 0.05. The exploration coefficient is 1.0. This is a training curriculum, not uniform sampling of the complete corpus.

The resident pool is rebuilt every 256 epochs. The full-scale run initially used 256 clips per rank and was resumed with 128 after memory failures; the local cache multiplier was reduced to 1.0. These changes affect residency and memory demand, not the underlying training split. Validation membership is fixed independently of pool rebuilding, and validation evaluation does not update training sampling weights. We report resident-pool and validation metrics separately because they measure different populations.

5 Experimental Setup

5.1 Development subset and full-scale protocol

The development corpus contains 150 clips and all 128 body configurations, for 19,200 variants. These clips are reserved within the full training split and are not a held-out generalization benchmark. Small-scale experiments investigate architecture and reference-processing choices with a fixed simulation budget. The full-scale run uses the 18,855 training clips in Table 1, with validation membership fixed throughout.

Table 3: Training configuration and budgets. Environment and minibatch counts are per GPU. The full-scale frame count is a recorded stopping snapshot, not a budget matched to the development runs.

Setting	Corrected 150-clip runs	Full-scale selected run
Parallel GPUs per policy	1	6
Environments per GPU	4,096	2,048
PPO minibatch size per GPU	16,384	8,192
Rollout length	32 control steps	32 control steps
Mini-epochs per update	1	1
Actor / critic learning rates	2×10^{-5} / 10^{-4}	2×10^{-5} / 10^{-4}
Discount / GAE / ratio clip	0.99 / 0.95 / 0.2	0.99 / 0.95 / 0.2
Gradient norm clip	50	50
Evaluation cadence	200 epochs	256 epochs
Training seed	0 (all eight runs)	Recorded in saved run configuration
Reference residency	All 150 clips	256, later 128 clips per GPU
Training frames	786,432,000 per run (6,000 epochs)	Approximately 11.07×10^9 recorded
Status of reported evidence	Completed, Table 8	Epoch-28,170 checkpoint, Table 7

The selected full-scale experiment is slot/type attention trained directly on refined references. Its code retains a historical `stage2` name, but this run does not use neutral pretraining followed by shape transfer. Table 3 records the shared optimization settings and the actual environment/batch sizes used after the initial memory-related launch adjustments. Run-specific resolved configurations take precedence over older launch examples.

5.2 Architecture and refinement controls

The corrected comparison is a four-architecture by two-reference-version grid, Table 4. The four architectures are a wide flat temporal MLP, basic temporal attention, attention with slot/type embeddings, and that last model plus actor-only adaLN-Zero. Each is trained once on refined references (A–D) and once on unrefined references (E–H); slot/type on unrefined references is labelled E rather than a separate letter for historical reasons. All eight runs use one seed, a single A40, the same 6,000-epoch budget, evaluation cadence, and shape-sampling seed.

All eight runs use seed 0 and one A40 GPU per policy. Morphology concatenation into both heads is retained in D and H; adaLN-Zero is additional actor conditioning, not a replacement for the critic or a two-stage schedule.

The models do not have equal parameter counts (Table 5). The grid is single-seed; the architecture ordering in Table 8 rests on the jerk difference, which recurs under both reference versions, and not on the success-rate column, which is treated as non-discriminating. The refined and unrefined pairs (A/F, B/G, C/E, D/H) are each evaluated against their own reference version, so they measure whether refined training costs policy quality, not whether it causally improves accuracy; the refinement argument is completed by the offline reference-artifact audit in Section 6.4 rather than by a common-reference controller re-evaluation. A causal comparison would require scoring refined- and unrefined-trained checkpoints on one shared reference version.

Earlier small-scale runs influenced the choice to begin full-scale slot/type training. The corrected reruns were launched later to reassess that choice; we do not present them as completed model selection preceding the full run.

Table 4: Corrected 150-clip grid: four architectures × two reference versions, one seed each. Results are in Table 8.

Case	References	Architecture	Comparison purpose
A	Refined	Wide temporal MLP; beta concatenation	Temporal MLP baseline.
B	Refined	Basic temporal attention	Attention versus flattened temporal input.
C	Refined	Attention with slot/type embeddings	Explicit temporal/source identity.
D	Refined	C + actor-only adaLN-Zero	Morphology modulation with critic unchanged.
E	Unrefined	As C	Reference version at fixed architecture (slot/type).
F	Unrefined	As A	Reference version at fixed architecture (temporal MLP).
G	Unrefined	As B	Reference version at fixed architecture (attention).
H	Unrefined	As D	Reference version at fixed architecture (slot/type + adaLN).

5.3 Morphology-matched evaluation and checkpoint scope

Each reference is rolled out only in an environment using its exact asset identity. A selected reference therefore cannot be assigned to a different body simply because an environment slot is available. Multiple rollout batches may be needed when several references require the same body.

Routine monitoring selects one variant per clip. With seed $s = 42$, stable clip identity c, panel index p, and N_c available variants, the choice is

$$k(c,p) = \big(\mathrm{uint64}_{\mathrm{big}}\big(\mathrm{SHA256}(s\!:\!c)_{0:8}\big) + p\big) \bmod N_c. \tag{9}$$

Here $s\!:\!c$ is the UTF-8 string formed as `seed:clip_id`. For scheduled monitoring, p is the integer quotient of the checkpointed epoch and evaluation cadence. This produces deterministic staggered shape choices that rotate across evaluations and remain reproducible after resume. The validation *clip list* is fixed; the selected *shape panel* changes. Thus successive scores are not repeated measurements on an identical set of clip/shape pairs.

Evaluation uses the mean action, reference initialization, and no gradient tracking. It disables training termination and records errors over the valid reference horizon, capped at 600 control steps (20 s at 30 Hz). Success therefore refers to the evaluated horizon, not necessarily the entire duration of a longer source clip.

Final model comparisons should freeze the checkpoint, clip/shape panel, reference version, simulator configuration, and seed. Validation is available for model selection and has been repeatedly monitored; it is not an untouched test set. The independent test manifest remains outside training, and no test result is reported in this draft. The downloaded epoch-28,170 `last.ckpt` is also distinct from the epoch-28,159 monitoring snapshot in Table 6; a new evaluation of that checkpoint must be reported separately.

5.4 Metrics and aggregation

For each evaluated clip/shape pair i, define its per-frame mean body position error

$$d_{i,t} = \frac{1}{B}\sum_{j=1}^{B} \|p_{i,t,j} - \hat{p}_{i,t,j}\|_2. \tag{10}$$

The configured success indicator is

$$S_i = \mathbf{1}\left[\max_{t<T_i} d_{i,t} \le 0.5 \text{ m}\right], \tag{11}$$

where T_i is the evaluated horizon. This threshold is deliberately distinct from the smoother training reward. Passing it does not imply fine-grained pose accuracy or biomechanical realism.

We also report time-averaged mean body-position error, mean global body-rotation geodesic error, and the time average of the largest body-position error within each frame. The last quantity is the logged *mean maximum-joint error*; it is not the maximum over the entire dataset. Global body-rotation error is not a local joint-angle/DOF error.

Within an evaluator rank, errors are averaged over valid frames per motion and then over motions, including failed rollouts. The distributed training logger aggregates scalar summaries across ranks; the monitoring table retains those logged values rather than reconstructing exact pooled counts. Final offline comparisons should aggregate saved per-motion records over unique clip/shape pairs, avoiding unequal weighting of partially filled batches or duplicate padded rollouts.

Trajectory smoothness uses the implementation's windowed normalized-jerk score, with the definition and numerical regularizers in Appendix A.3. Lower scores indicate lower jerk under that fixed convention. The accompanying high-jerk statistic is the percentage of windows in which any body exceeds threshold 6,500; the logger calls it a frame percentage, but it is window-based.

Action increments require care. The current evaluator stores raw policy actions, before Equation (3). Their differences are policy-output changes, not physical joint-angle changes, even where logger field names label them radians or degrees. We omit those mislabeled angular values from the physical interpretation and from Table 6. PD-target changes, applied torques, and measured joint motion must be computed separately.

5.5 Generalization axes and compute accounting

Held-out clips on trained bodies, seen clips on unseen bodies, and held-out clips on unseen bodies are different evaluations. The current validation snapshot addresses only the first. A single rotating panel does not measure all 128 variants of every clip. Future cross-shape comparisons must report the actual body identities and number of evaluated pairs. The planned unseen-morphology evaluation in Section 6.5.1 targets the third axis directly, on shapes drawn from the trained coefficient range but excluded from the 128 training configurations, with results stratified by distance to the nearest training morphology.

We retain W&B logs, resolved configurations, checkpoint epochs, and local results for compute accounting. Small-scale development and full-scale training costs should be reported separately, including resumed segments and failed launches when discussing practical cost. GPU-hours, rather than epoch counts alone, quantify this budget.

Table 5 reports parameter counts and single-A40 wall-clock cost for the eight corrected 150-clip runs, all completed at the 6,000-epoch budget. Parameter counts are computed analytically from each run's resolved model configuration (encoder, transformer or flat-MLP, and head widths), not read from a stored checkpoint file, since every linear layer in this codebase is instantiated with `nn.LazyLinear` and has no fixed parameter count outside a materialized model. Parameter count is set by the architecture only, so the refined/unrefined pairs share a count (A/F, B/G, C/E, D/H).

Two consequences run against a naive intuition. First, the temporal MLP has the most total parameters of any architecture, not the fewest: its actor's first layer reads the full raw concatenation of every observation category (5,248 dimensions) directly, which alone costs more parameters

Table 5: Parameter counts and single-GPU wall-clock cost for the eight corrected 150-clip runs (seed 0, one A40 each, 6,000 epochs). Actor and critic are separate networks with independent parameters.

Case	Arch	Actor params	Critic params	Total params	GPU-h
A / F	temporal MLP	57,349,557	8,524,801	65,874,358	15.8 / 14.4
B / G	attention	45,430,197	5,800,705	51,230,902	23.5 / 21.5
C / E	slot/type	45,434,037	5,804,545	51,238,582	23.7 / 23.9
D / H	slot/type + adaLN	46,256,053	5,804,545	52,060,598	23.9 / 22.4

than the attention models' three token encoders and transformer combined. The attention models compress that same information to a 256-dimensional token per source before the shared final head, trading parameters for the added transformer computation. Second, despite having more parameters, the temporal MLP is the fastest to train by a wide margin (15.8/14.4 vs. 21.5–23.9 GPU-hours) – parameter count does not predict wall-clock cost here, since the attention models pay for multiple sequential module calls (three encoders, then a transformer stack) rather than one large matrix multiplication chain. Actor-only adaLN-Zero adds a small but non-negligible cost over slot/type (roughly 822,000 actor parameters from the per-layer modulation networks and morphology encoder), consistent with, but not sufficient by itself to explain, the jerk difference in Section 6.3.

The full-scale selected run shares C and E's architecture exactly (same encoder, transformer, and head widths; `mlp_wide_stage2_discover_attention_slot_type.py` inherits these constants unchanged from the 150-clip experiment file), so its parameter count is also 51,238,582. Its own GPU-hours are not included here: it ran on six GPUs with resumed segments and a changed resident-pool size after a memory-related failure, which is not a single comparable wall-clock number in the way the eight matched single-GPU development runs are.

Pending evidence. Add a resumed-run GPU-hour accounting for the full-scale training, including the failed launch before the resident-pool-size correction. No superior compute-efficiency claim is made from the current evidence – the temporal MLP's lower wall-clock cost is not weighed against its higher position error relative to slot/type in Table 8.

6 Results and Failure Analysis

6.1 Recorded full-scale validation result

Table 6 reports the final scheduled monitoring evaluation recorded for the selected full-scale slot/type run, at epoch 28,159 after approximately 11.07 billion simulated frames. The separately downloaded checkpoint is from epoch 28,170. These values are post-correction, morphology-matched evaluations, not measurements of that later checkpoint or of the independent test split.

The validation score is higher than the resident-pool score, while the reported errors are lower. This comparison is consistent with a curriculum that concentrates residency on difficult clips: the resident pool is not an independent uniform sample of the training distribution. Its 89.84% success rate should not be interpreted as the success rate of the entire training corpus, nor should the difference be described as validation outperforming an IID training sample.

The 98.47% result supports strong tracking of the selected held-out clip/body pairs under the specified threshold. It does not establish all-body success for those clips, unseen-body generalization, or an untouched test-set score. The thresholded success rate is complemented by position,

Table 6: Recorded full-scale monitoring snapshot. Validation contains 1,048 held-out clips; each evaluation uses one selected trained body per clip. The active resident pool is difficulty-biased. Values are the stored distributed logging summaries from run `3mvgad6f`, epoch 28,159; no confidence intervals or all-shape coverage are implied.

Metric	Hard resident pool	Validation holdout
Success rate (%) ↑	89.84	98.47
Mean body-position error (m) ↓	0.0960	0.0698
Mean global body-rotation error (rad) ↓	0.2261	0.1760
Mean maximum-joint position error (m) ↓	0.1848	0.1412
Windowed normalized jerk ↓	1,442.8	1,151.7
High-jerk windows (%) ↓	31.05	27.25

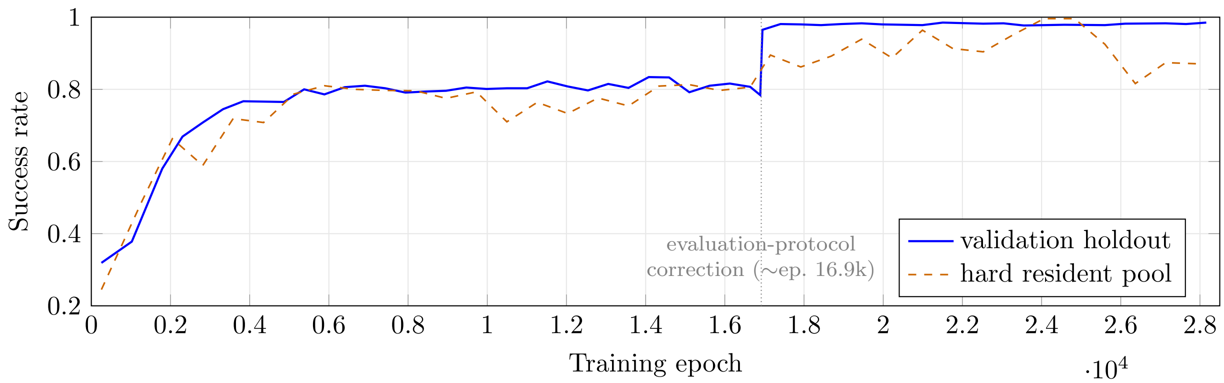

Figure 3: Scheduled success rate over training for the full-scale slot/type run (`3mvgad6f`). The step near epoch 16.9k is the morphology-matched evaluation correction plus its sampling change, not an instantaneous learning event; pre-correction points use the earlier, sometimes mismatched reference-to-asset assignment and are not comparable to the post-correction curve. After the correction the holdout success rate climbs from about 0.96 to a stable 0.98–0.985. The resident-pool curve is noisier because residency is continually re-concentrated on hard clips.

rotation, and smoothness measures because a successful rollout can still have visible local errors or jitter.

The run record is available at the full-scale W&B run. The current table uses the recorded snapshot, not a newly executed evaluation.

6.2 Standalone validation and held-out test evaluation

The epoch-28,170 checkpoint was re-evaluated offline with a standalone streaming evaluator, once on the validation split and once on the independent test split. Both runs use the same protocol as training-time monitoring – one deterministically selected body shape per clip, an environment with that exact asset, the mean action, and a capped horizon – and write a per-motion record for every clip. The test manifest is loaded only by this standalone evaluator; its SHA-256 and clip count are checked against the split metadata, and its disjointness from train and validation is verified before evaluation. All test clips are the standard 6.6 s HumanML3D crop, so none reach the horizon cap. The test split is evaluated once; its number is not iterated against.

Test performance is marginally better than validation on every metric (Table 7). This is the

Table 7: Offline evaluation of the epoch-28,170 checkpoint on the 1,048-clip validation and 1,048-clip test splits of `hhi_stage2_v1`, one selected trained body per clip (rotating shape panel index 110). Same standalone evaluator for both.

Metric	Validation	Test
Success rate (%) ↑	98.47	**99.14**
Failure rate (gt err > 0.5) (%) ↓	1.53	**0.86**
Mean body-position error (m) ↓	0.0699	**0.0676**
Mean global body-rotation error (rad) ↓	0.1738	**0.1723**
Mean maximum-joint position error (m) ↓	0.1419	**0.1393**
Normalized jerk ↓	1,154.3	**1,050.9**
High-jerk frames (%) ↓	27.46	**25.33**
Worst single clip, gt err max (m)	3.79	**1.67**

Table 8: Corrected 150-clip comparison, last-8-evaluation mean ± standard deviation. Refined-data cases (A–D) are scored against refined references and unrefined-data cases (E–H) against unrefined references; the two halves are not a controlled cross-comparison (Section 6.4). Parameter counts and GPU-hours are from Table 5. Single seed per case.

Case	Architecture / data	Success (%)	gt err	gr err	Jerk	HJ%
A	temporal MLP, refined	97.5 ± 0.8	0.109	0.207	1,729	33.3
B	attention, refined	97.5 ± 1.4	0.098	0.228	3,455	40.4
C	slot/type, refined	**98.1 ± 0.6**	0.097	0.226	**1,757**	34.2
D	slot/type + AdaLN, refined	97.1 ± 1.2	**0.090**	0.220	3,145	38.9
E	slot/type, unrefined	97.9 ± 0.6	0.102	0.219	1,422	29.6
F	temporal MLP, unrefined	96.2 ± 1.6	0.115	0.212	2,927	36.6
G	attention, unrefined	97.8 ± 1.2	0.098	0.219	3,117	34.9
H	slot/type + AdaLN, unrefined	97.5 ± 0.9	0.091	0.207	2,956	34.0

expected outcome for a held-out reveal: validation was monitored during development while test was not, and the roughly 0.7-point success gap is within the variation expected between two random 5% partitions of the same corpus. The result does not establish all-body coverage for these clips, unseen-body generalization, or any claim outside the rotating one-shape-per-clip protocol.

Pending evidence. Add a frozen shape-panel repeat and explicit all-panel coverage for a subset. Per-motion records for both splits are retained but not yet released.

6.3 Corrected architecture comparison

An earlier evaluator could assign a reference to an environment with the wrong morphology. Its correction changed the meaning of the evaluation: a rise at the code transition cannot be attributed to instantaneous policy learning. The accompanying sampling correction also changed which training clips entered the resident pool. These changes must be distinguished from an architectural improvement.

The eight corrected runs in Table 4 reassess the 150-clip comparisons using the matching-asset protocol, each for the full 6,000-epoch budget with one training seed. Table 8 reports the mean and standard deviation of each run's last eight scheduled evaluations (epochs approximately 4,600–6,000); single evaluation points swing by two to four percentage points of success rate and by 100–300 in jerk, so a single final snapshot is not a stable summary.

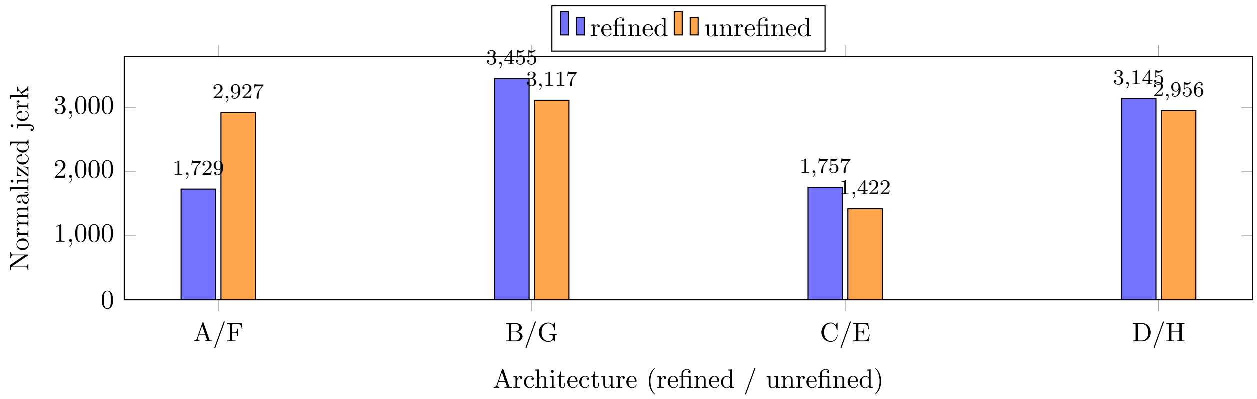

Figure 4: Normalized jerk (last-8-evaluation mean) for the eight 150-clip runs, grouped by architecture. Basic attention (B/G) and slot/type + adaLN-Zero (D/H) roughly double the jerk of the plain temporal MLP (A/F) and of slot/type without adaLN-Zero (C/E), on both reference versions. Success rate is 96–98% for all eight (Table 8) and does not distinguish them.

Success rate does not separate the runs: all eight sit between 96 and 98% with last-8 standard deviations of 0.6–1.6 points, and the intervals overlap. Jerk, and the high-jerk-window percentage that tracks it, is the one axis that separates them consistently (Figure 4). On the clean within-data comparison (A–D, identical refined targets), the slot/type run (C) has the highest success rate and low jerk; basic attention (B) and actor-only adaLN-Zero (D) both roughly double the jerk of C, and D's slightly lower position error is within one standard deviation while its jerk and high-jerk fraction are clearly worse. The temporal MLP (A) matches C on jerk but has clearly higher position error, and is the largest model by parameter count despite training fastest (Table 5). The unrefined group (E–H) reproduces the same ordering: slot/type (E) has the lowest jerk of the four, and adding adaLN-Zero (H) or removing the slot/type embeddings (G) roughly doubles it.

These are single-seed development results and the pre-correction 150-clip curves are not comparable to them. The load-bearing signal is the jerk difference, which is a factor-of-two effect and appears in the same ordering under both reference versions – basic attention and adaLN-Zero high, plain MLP and slot/type low – so it is a within-experiment replication rather than one lucky run; the success-rate column, by contrast, is treated as non-discriminating. Within those limits the grid supports the selected configuration, slot/type attention without adaLN-Zero, as the best-balanced choice rather than merely the incumbent.

Pending evidence. Report the explicit-panel cross-shape numbers for the selected configuration. Seed repeats would tighten the success-rate column but are not expected to move the jerk-based ordering.

6.4 Reference refinement: quality and policy cost

Refinement is justified on a deliberately narrow basis: it removes measurable artifacts from the reference motion itself, and training on the refined references does not cost policy quality. We do not claim a causal accuracy benefit from refined training, which the present evidence cannot establish. The two parts of the argument are evaluated separately.

The reference-level part is an offline before/after audit on the packaged motion tensors, with no simulation, over all 19,200 references of the 150-clip development set (150 clips $\times$ 128 body configurations). Table 9 reports the artifact measures – stance-foot horizontal speed during detected contact, the fraction of frames whose lowest body centre is below the floor, and reference-domain

Table 9: Reference-refinement audit over all 19,200 development references. Artifact rows: corpus mean before and after, and the fraction of individual motions that improve; the per-motion distributions are in Figure 5. Fidelity rows: corpus mean of the refined-vs-unrefined difference. Jerk is the raw mean third-difference of body positions, not the windowed evaluator score.

	Unref. mean	Refined mean	Motions improved
Stance-foot speed (m/s) ↓	0.055	0.038	96%
Frames below floor (%) ↓	39.6	3.3	45%
Frames within 2 cm of floor (%) ↓	49.8	33.6	—
Position jerk (m/s^3)	45.2	60.7	21%
Per-body rotation change (deg)		0.12	
Root horizontal displacement (m)		0.008	
Cross-shape var., knee-flexion range (ratio r/u)		0.94	

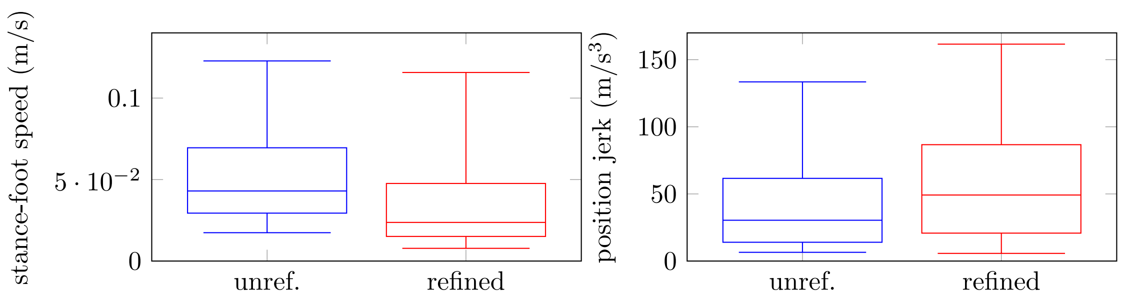

Figure 5: Per-motion distributions (box: quartiles; whiskers: 5th–95th percentile) over the 19,200 development references. Foot-skate drops across almost the whole distribution; position jerk rises across almost the whole distribution.

position jerk – and the fidelity measures – per-body geodesic rotation change, root horizontal displacement, and the cross-shape variance of per-shape knee-flexion range, which must be preserved rather than flattened.

Refinement removes most ground penetration and about a third of the stance-foot sliding, while changing joint rotations by a tenth of a degree and the root path by millimetres, and leaving the cross-shape spread of per-shape knee kinematics essentially intact (variance ratio 0.94). The per-motion distributions (Figure 5) show these are not mean-only effects: stance-foot speed falls for 96% of individual motions, across the full range rather than a few outliers, although the worst-sliding 5% of clips are barely improved. Penetration in the raw references is bimodal – a large fraction of clips sit entirely below the floor, a whole-clip vertical mis-registration rather than per-frame jitter, which the clearance correction removes; clips that were already grounded stay grounded, so only 45% of motions register a change. The cost is a roughly one-third increase in reference-domain position jerk that is broad rather than tail-driven (79% of motions get jerkier, and the per-shape medians move together, 45.5 to 60.5), introduced by the per-window root-translation correction and the per-frame ground-clearance lift. This also explains the earlier observation that the unrefined slot/type run (E) shows lower policy-side jerk than the refined one (C): the refined *reference* itself carries more position jerk, so a policy that tracks it faithfully inherits more, and the E–C gap reflects the target, not a smoother policy from unrefined training.

The policy-cost part reuses the architecture-matched refined and unrefined training pairs from Table 4 (A/F, B/G, C/E, D/H). Each run is evaluated against its own reference version, so these pairs are not a controlled measurement of a refined-trained policy being more accurate; they show only whether refined training loses anything. Across the four pairs the differences are small and mostly within single-seed noise: the refined run has the lower mean position error in three of four pairs, success rate is within half a percentage point either way in three pairs and clearly higher for the refined run in the temporal-MLP pair, and the jerk difference has inconsistent sign once the reference-jerk offset above is taken into account.

Refined references are therefore used for the full-scale run because they remove most ground penetration and much of the foot sliding, with negligible change to joint angles or root trajectory and no loss of cross-shape variation, and at no measurable policy-accuracy cost. The added reference-domain jerk is a known trade-off. A stronger, causal claim would require evaluating refined- and unrefined-trained checkpoints on one common reference version.

Pending evidence. Add per-clip and per-shape distributions for Table 9 rather than corpus means. No causal full-scale refinement benefit is claimed.

6.5 Cross-shape consistency and qualitative evidence

A useful qualitative comparison places several bodies performing the same clip side by side, with reference markers and a shared time axis. Source descriptions can identify the action, provided mirrored and cropped captions have been checked. Reference-only videos illustrate the generated corpus but must not be presented as successful policy rollouts.

For quantitative cross-shape analysis, the same clip should be evaluated on an explicit body panel. Report both aggregate accuracy and variation across shapes, including the number of clips that succeed on every tested body. A rotating one-shape snapshot does not provide this statistic. Unseen beta vectors require corresponding new assets and references and must be evaluated separately from trained bodies, even if their coefficients lie within the same $[-3, 3]$ range.

6.5.1 Planned unseen-morphology evaluation

Training fixes 128 body configurations sampled inside $\boldsymbol{\beta} \in [-3, 3]^{10}$ (Section 3.2). To test whether the controller has learned morphology-conditioned control rather than memorized those 128 configurations, we plan a held-out-body evaluation on approximately 20,000 motion clips whose body shapes are *not* among the 128 training combinations but lie within the same coefficient range. New MJCF assets and HUMOS references are generated for these shapes exactly as in Section 3.2, and rollouts follow the Section 5.4 protocol.

The primary readout is success rate. A value that stays near the trained-body level would indicate that conditioning, not enumeration, drives the behavior; a large drop would indicate memorization of the training morphologies. The sharper analysis plots each component error and the success indicator against the Euclidean distance from the evaluated $\boldsymbol{\beta}$ to its nearest training morphology in coefficient space. Performance that stays approximately flat as this distance grows is the strong form of the claim; a monotone decline with distance would bound generalization to a neighborhood of the training set. Any resulting statement is scoped to *generalization to unseen morphologies within the trained coefficient range*, not to arbitrary body shapes, and unseen bodies are never mixed into the trained-body success numbers.

Pending evidence. Generate the held-out-shape assets and references, run the distance-stratified evaluation, and report success and component errors as a function of distance to the nearest training morphology, with the trained-body and unseen-body populations kept separate. Insert matched reference/controller visualizations and the explicit-panel cross-shape results. Joint-space generalization remains unmeasured in the evidence reported here.

Table 10: Residual-failure analysis framework. Categories are hypotheses until supported by measurements; they can overlap and include an unresolved group.

Candidate explanation	Evidence required
Reference artifact	Sliding, penetration, abrupt rotation/root changes, or inconsistent contacts at the failure time; compare refined and original references.
Missing environmental support	Reference/video and checked caption show an object, seat, wall, partner, or other support absent from the simulated task.
Morphology-specific difficulty	The same clip succeeds on other bodies; compare geometry, clearance, joint limits, and reference differences on the failing body.
Controller or optimization limitation	Repeated failure despite an apparently valid reference; recovery under additional targeted training or an independently tested controller would argue against intrinsic infeasibility.
Metric or protocol effect	Inspect asset identity, horizon, initialization, and threshold crossings; check whether a visually plausible deviation alone changes the binary label.
Unresolved	Retain ambiguous and mixed cases without assigning a physical cause.

6.6 Evidence-based analysis of residual failures

We do not assume a fixed 10% failure tail or an intrinsic 80% learning ceiling. The observed failure fraction depends on the evaluated population, shape panel, checkpoint, horizon, and threshold. Moreover, the old morphology mismatch produced false failures. A residual failure should therefore be treated as an observation to explain, not evidence that the motion is physically impossible.

The analysis unit is a failed clip/body pair with a reproducible rollout. For each pair, retain the source identity, morphology, first threshold crossing, tracking-error trajectory, and a synchronized reference/controller visualization. Compare it with a successful body variant of the same clip where available. Table 10 defines candidate explanations and the evidence needed to support them.

A complete report should quantify these categories over all evaluated failures, or over a documented stratified sample with appropriate denominators, and include representative successes as controls. Repeated rollouts and targeted interventions can strengthen an explanation; visual inspection alone cannot prove dynamic infeasibility. Capacity limits, reference defects, and body-dependent dynamics may coexist.

6.6.1 Failures on the validation and test splits

Joining the 9 test failures and 16 validation failures (Table 7) to their inherited HumanML3D captions gives a consistent picture across both splits. Every failed clip is a recognizably hard motion; none is an unexplained failure of an ordinary movement. Two motion families account for most of the 25:

- *Transitions to and from the ground* – getting up from lying or prone, belly crawling, push-up postures, sit-to-stand-to-floor: 6 of 9 test failures and 5 of 16 validation failures.
- *Whole-body dynamic rotation* – cartwheels, a running somersault: 2 of 9 test and 4 of 16 validation.

The remainder are turning or side-stepping locomotion (5 validation), one clip whose caption describes a hand-held sparring stick that the flat-ground task does not simulate (validation `M006386`, the single clear missing-support case), one abrupt run-to-stop, and one test clip (`M013267`) whose peak position error is exactly at the 0.5 m threshold and whose caption is garbled, which we treat as a metric-boundary case rather than a tracking failure. Several base motions fail on both their original and mirrored orientation, each on a different body shape (three base motions in test, about six in validation), which is consistent with a skill-class difficulty that interacts with morphology rather than with a single unlucky clip. The ground-transition and acrobatic clusters coincide with the historically hard skill classes seen throughout the development lineage, which points to a controller or optimization limitation on those classes; the acrobatic failures also carry very high policy-side jerk, and since the corpus-level audit in Section 6.4 shows refinement raises reference jerk on average, a reference artifact on those specific clips remains a live possibility pending a per-clip check.

Pending evidence. Cross-check the acrobatic-cluster references against the refinement audit, and confirm the both-orientation failures against successful shapes of the same clip on an explicit panel. No single causal attribution is claimed for the mixed cluster.

7 Physical Analysis

Planned analysis; no results in this draft.

This section will examine how body morphology changes the physical control demand of a motion, using the simulator's instrumented quantities and not claims about human biomechanics. The question is not whether body mass or size correlates with joint torque, which is partly expected from mechanics, but whether the effect of morphology on control demand *depends on the motion being performed.*

The planned procedure holds the motion fixed and varies the body. For a matched set of clips evaluated across many of the 128 configurations, we record per-joint torque, joint power, an aggregate control-effort measure, and jerk from the simulator during deterministic rollouts. Clips are then grouped by their inherited text descriptions into coarse semantic classes (for example walking, running, jumping, kicking, dancing, sitting), and we compare the cross-morphology variation of each quantity between classes.

The hypothesis under test is that morphological variation produces motion-dependent changes in control demand, with highly dynamic motions (running, jumping, kicking) more sensitive to body shape than slow or quasi-static motions (standing, sitting) – that is, body morphology and motion characteristics interact in setting control demand, rather than morphology acting as a motion-independent scaling. A concise result would state that relationship together with per-class distributions, not a single correlation coefficient.

Joint-torque and contact-force readouts, their normalization across bodies of different mass and size, and the simulator instrumentation must be validated before any of these quantities are reported.

Pending evidence. Define the matched clip set and the semantic grouping, validate the torque, power, effort, and jerk instrumentation and its cross-body normalization, and report the morphology $\times$ motion-class interaction with per-class distributions. No mechanical or biomechanical claim is made in this draft.

8 Discussion and Limitations

The current evidence supports the feasibility of combining shape-specific references, matching simulation assets, and one shared controller at the scale of the constructed corpus. It does not yet identify the best architecture or establish a causal learning benefit from refinement. Those conclusions depend on the corrected comparisons, not on the pre-correction development curves.

The dataset is derived from existing motion capture, captions, body models, and HUMOS generation. The 2,681,728 variants represent combinations of 20,951 clip identities and 128 configurations, not equally many independent human examples. Mirrored clips are also not independent captures. Caption alignment, left/right semantics, body-specific wording, and redistribution permissions require further auditing before a benchmark release or language-grounded performance claim.

Refinement changes kinematics rather than solving the complete equations of motion. Ground clearance of collision proxies does not guarantee balanced support, plausible contact forces, or realistic joint torques. Similarly, estimated asset masses and mass-scaled actuators are modeling choices rather than a calibrated description of human physiology. The flat-ground task excludes much environmental interaction, and text-based filtering cannot ensure that every remaining clip satisfies that scope.

Generalization is limited by the split and evaluation design. Original and mirrored variants stay together, but the current split is not demonstrated to be subject- or source-recording-disjoint. Validation reuses the trained body set and has been monitored during development. Its rotating panel covers only one shape per clip at a time, and long clips have a capped evaluation horizon. Fixed-pair test evaluation and the unseen-morphology experiment in Section 6.5.1 remain necessary for broader claims, and any morphology-generalization statement is confined to body shapes inside the trained coefficient range rather than arbitrary bodies.

The success threshold is permissive relative to fine-grained pose accuracy. A high success rate should be read together with continuous errors, smoothness, and qualitative rollouts. The logged normalized-jerk measure also has implementation-specific windows and regularizers; it is not a standalone certificate of physical plausibility. Raw policy-action increments must not be interpreted as joint-angle or torque measurements.

Cost constraints motivated small-scale development and streamed full-scale training, and introduced memory-related resumes and configuration changes on the full-scale run. The 150-clip grid is single-seed: the architecture conclusion is drawn from the jerk difference, which is a large effect that recurs across both reference versions, and not from the success-rate differences, which are within run-to-run noise and are reported as non-discriminating. Seed repeats would sharpen the latter but are not expected to change the former. These costs and choices should be reported transparently; they do not by themselves establish better compute efficiency than another method.

We explored neutral-body pretraining followed by multi-shape transfer and mixed canonical/HUMOS source-weighted rewards during development, but did not retain either as the main approach. Their historical measurements are not used as controlled negative results, particularly where the earlier evaluation protocol differed. The present method is direct multi-shape training. Future work may revisit these alternatives after the current comparisons and dataset audits are complete.

9 Conclusion

We have described a text-associated multi-morphology motion corpus and a physics-based imitation system that preserves the correspondence between source clips, generated references, and simulated bodies. The construction combines AMASS/HumanML3D provenance, HUMOS generation, SMPLSim assets, conservative reference refinement, and deterministic clip-grouped splits. The controller uses temporal slot/type attention with morphology information in both actor and critic and is trained directly across body configurations.

Offline evaluation of the selected checkpoint gives 98.47% success on validation and 99.14% on the untouched test split for the one-shape-per-clip trained-body protocol, and the eight-run development grid selects slot/type attention without adaLN-Zero. Residual failures on both splits are dominated by ground-contact transitions and whole-body acrobatics. These results do not establish all-shape coverage or unseen-body generalization. Two studies are specified but not run: an unseen-morphology evaluation over body shapes inside the trained coefficient range, and an analysis of how morphology and motion type jointly set physical control demand – whether dynamic motions are more shape-sensitive than quasi-static ones.

A Reproducibility Details

A.1 Source preparation and refinement settings

The AMASS source collections used by the preparation pipeline are ACCAD, BioMotionLab_NTroje, BMLhandball, BMLmovi, CMU, DFaust_67, EKUT, Eyes_Japan_Dataset, HumanEva, KIT, MPI_HDM05, MPI_Limits, MPI_mosh, SFU, SSM_synced, TCD_handMocap, TotalCapture, and Transitions_mocap. Source preparation operates at 20 Hz; the final motion libraries use 30 Hz. Joint ordering and quaternion conventions must follow the matching asset and MotionLib conversion rather than be inferred from source array positions.

Table 11: Default settings of the current reference-refinement procedure. Frame counts refer to the 30 Hz MotionLib representation. These are kinematic-processing settings, not physical constraints learned by the policy.

Operation	Setting
Contact candidates	Left/right ankle and toe bodies
Contact detection thresholds	Speed 0.15 m/s; height 0.10 m
Contact cleanup	Close gaps up to 2 frames; remove runs shorter than 3 frames
Rotation filter	Savitzky–Golay, 9-frame window, polynomial order 3
Rotation blend / bound	Strength 0.5; maximum geodesic correction 8°
Contact-transition guard	Separate segments; 2-frame transition taper
Root XY correction	Median stance anchors; least-squares common translation
Root correction regularization	21-frame smoothing window; 0.10 m displacement cap
Ground clearance	0.005 m collision-geometry margin; 5-frame lift smoothing
Final reconstruction	Per-asset FK and per-motion velocity recomputation

A.2 PD gains and limits

Table 12 gives the base values configured for the SMPL morphology robot. For joint d belonging to rigid body j, the simulator applies the multiplier

$$\rho_{b,j} = \frac{m_{b,j}}{m_{\mathrm{ref},j}} \tag{12}$$

to stiffness, damping, and effort limit. The velocity limit is not mass-scaled. All relevant asset masses, joint ranges, collision settings, and the reference asset must accompany a reproducible physical interpretation.

Table 12: Base PD and actuation values before per-body mass scaling. Stiffness and damping use the simulator's rotational PD units (Nm/rad and Nm s/rad); effort limits are in Nm and velocity limits in rad/s.

Joint family	Stiffness	Damping	Effort limit	Velocity limit
Hip, knee, ankle	800	80	500	100
Toe	500	50	500	100
Torso, spine, chest	1,000	100	500	100
Neck, head, thorax, shoulder, elbow	500	50	500	100
Wrist, hand	300	30	500	100

A.3 Windowed jerk convention

The logged smoothness score is computed from simulated global body positions using consecutive finite differences with control interval $\Delta t = 1/30\,\mathrm{s}$. A nominal 0.4 s window contains $W = \max(4, \lfloor 0.4/\Delta t \rfloor)$ frames, giving $W = 12$ at the configured rate and duration $T_w = (W-1)\Delta t$. For body j in window w, let $L_{w,j}$ be the sum of speed times Δt and let $\dddot{p}_{w,j}$ denote the third finite difference. The implementation uses

$$J_{w,j} = \frac{T_w^5 \sum_{t \in w} \| \dddot{p}_{t,j} \|_2^2 \, \Delta t}{\max(L_{w,j}, \epsilon)^2 + \epsilon}, \qquad \epsilon = 0.1, \tag{13}$$

where the sum includes the valid third-difference samples in the window. Window scores are averaged per body and then across bodies and motions. High-jerk percentage counts windows for which $\max_j J_{w,j} > 6{,}500$.

Equation (13) documents the existing numerical convention, including its regularizers; it is not presented as a universally dimensionless or sampling-rate-invariant biomechanical measure. Comparisons must keep the sample rate, window, units, regularizers, and aggregation fixed. Reference refinement reports also include raw jerk in m/s^3; those values are a different metric and cannot be compared numerically to this logged score.

A.4 Split and experiment provenance

The split metadata is stored in `data/splits/hhi_stage2_v1/split_metadata.json`, with schema version 1, split seed 42, and split identifier

```
4c1c9d592e3dbeef15abe18cb50a3141820735849414ecc2d8959e9f38502263
```

It records the source-manifest hash, training/validation/test manifest and ID-list hashes, difficulty information, and the forced training reservation. The generator uses 100 difficulty strata; the 150 requested development identities expand to 286 retained identities when existing mirrored partners are included. Checkpoint/run split provenance refers to these artifacts, not just a verbal 90/5/5 ratio.

The documented HUMOS configuration selects `logs/humos/q6zbv2tu/checkpoints/latest-epoch=1599.ckpt`. Exact HUMOS and asset-generation revisions and checkpoint checksums must be archived with the final dataset specification. A checkpoint path identifies the local workflow but is not a substitute for an immutable checksum.

The principal implementation entry points are:

- Reference refinement: `tools/refine_humos_motion.py`; streamed per-clip processing: `tools/refine_humos_r2_per_clip.py`.

- Selected full-scale experiment: `examples/experiments/mimic/mlp_wide_stage2_discover_attention_slot_type.py`, inheriting the common attention experiment and enabling slot/type embeddings.

- Corrected development comparison: `tools/train_150_evalfix_ablation.sh`, cases A–E with seed 0.

- Motion residency and evaluation: `protomotions/components/global_clip_pool.py` and `protomotions/agents/evaluators/mimic_evaluator.py`.

- Standalone validation: `protomotions/evaluate_validation_split.py`. Its output must record the selected checkpoint, split, reference version, and shape panel.

For each final experiment, retain the resolved configuration, code revision, model/checkpoint checksum, body coefficient and asset manifests, split hashes, all resume segments, frame count, GPU model/count, and elapsed GPU-hours. For each evaluation, retain per-pair clip and asset IDs, panel selection, success/failure, valid horizon, and component errors in a local analysis file. The main full-scale W&B identifier is `3mvgad6f`; the corrected small-scale group is `hhi_150_evalfix_comparison`. The evaluation/sampling correction is recorded in commit `3685247`.

Pending evidence. Complete the immutable external-model/asset provenance, parameter counts, GPU-hour accounting, final per-pair evaluation exports, and caption coverage/alignment audit before treating this draft as a finalized benchmark report.

References

[1] Naureen Mahmood, Nima Ghorbani, Nikolaus F. Troje, Gerard Pons-Moll, and Michael J. Black. AMASS: Archive of Motion Capture as Surface Shapes. In *ICCV*, 2019. https://arxiv.org/abs/1904.03278.

[2] Chuan Guo, Shihao Zou, Xinxin Zuo, Sen Wang, Wei Ji, Xingyu Li, and Li Cheng. Generating Diverse and Natural 3D Human Motions From Text. In *CVPR*, 2022. CVF Open Access paper.

[3] Shashank Tripathi, Omid Taheri, Christoph Lassner, Michael J. Black, Daniel Holden, and Carsten Stoll. HUMOS: Human Motion Model Conditioned on Body Shape. In *ECCV*, 2024. https://arxiv.org/abs/2409.03944.

[4] Xue Bin Peng, Pieter Abbeel, Sergey Levine, and Michiel van de Panne. DeepMimic: Example-Guided Deep Reinforcement Learning of Physics-Based Character Skills. *ACM Transactions on Graphics*, 37(4), 2018. https://xbpeng.github.io/projects/DeepMimic/index.html.

[5] Zhengyi Luo, Jinkun Cao, Alexander W. Winkler, Kris Kitani, and Weipeng Xu. Perpetual Humanoid Control for Real-time Simulated Avatars. In *ICCV*, 2023. CVF Open Access paper.

[6] The ProtoMotions contributors. ProtoMotions. Open-source software for physics-based character animation and humanoid control. https://github.com/NVlabs/ProtoMotions.

[7] Viktor Makoviychuk, Lukasz Wawrzyniak, Yunrong Guo, Michelle Lu, Kier Storey, Miles Macklin, David Hoeller, Nikita Rudin, Arthur Allshire, Ankur Handa, and Gavriel State. Isaac Gym: High Performance GPU-Based Physics Simulation For Robot Learning. Technical report, 2021. https://arxiv.org/abs/2108.10470.

[8] John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. Proximal Policy Optimization Algorithms. Technical report, 2017. https://arxiv.org/abs/1707.06347.

[9] Jungdam Won and Jehee Lee. Learning Body Shape Variation in Physics-based Characters. *ACM Transactions on Graphics*, 38(6), Article 207, 2019. https://doi.org/10.1145/3355089.3356499.

[10] Matthew Loper, Naureen Mahmood, Javier Romero, Gerard Pons-Moll, and Michael J. Black. SMPL: A Skinned Multi-Person Linear Model. *ACM Transactions on Graphics*, 34(6), 2015. https://smpl.is.tue.mpg.de/.

[11] Zhengyi Luo and contributors. SMPLSim. Open-source SMPL-family body-model simulation software. https://github.com/ZhengyiLuo/SMPLSim.

[12] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, and Illia Polosukhin. Attention Is All You Need. In *NeurIPS*, 2017. https://arxiv.org/abs/1706.03762.

[13] William Peebles and Saining Xie. Scalable Diffusion Models with Transformers. In *ICCV*, 2023. https://arxiv.org/abs/2212.09748.

[14] Zhengyi Luo, Jinkun Cao, Josh Merel, Alexander Winkler, Jing Huang, Kris Kitani, and Weipeng Xu. Universal Humanoid Motion Representations for Physics-Based Control. In *ICLR*, 2024. https://arxiv.org/abs/2310.04582.

[15] Tairan He, Zhengyi Luo, Xialin He, Wenli Xiao, Chong Zhang, Weinan Zhang, Kris Kitani, Changliu Liu, and Guanya Shi. OmniH2O: Universal and Dexterous Human-to-Humanoid Whole-Body Teleoperation and Learning. In *CoRL*, 2024. https://arxiv.org/abs/2406.08858.

[16] Mazeyu Ji, Xuanbin Peng, Fangchen Liu, Jialong Li, Ge Yang, Xuxin Cheng, and Xiaolong Wang. ExBody2: Advanced Expressive Humanoid Whole-Body Control. Technical report, 2024. https://arxiv.org/abs/2412.13196.

[17] Zixuan Chen, Mazeyu Ji, Xuxin Cheng, Xuanbin Peng, Xue Bin Peng, and Xiaolong Wang. GMT: General Motion Tracking for Humanoid Whole-Body Control. Technical report, 2025. https://arxiv.org/abs/2506.14770.